THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS · MEDIUM LEVEL FULL-YEAR PAPER SET 2

Full Syllabus (Chapters 1–8) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. Two interior angles on the same side of a transversal cutting parallel lines are in the ratio $1:2$. The smaller angle is: 1

(a) 70° **(b)** 60° **(c)** 80° **(d)** 50°

2. Applying Kaprekar's routine once to 4172 gives: 1

(a) 6173 **(b)** 6176 **(c)** 6174 **(d)** 6175

3. The exterior angle at B of $\triangle ABC$ is 120° and $\angle A = 45^\circ$. Then $\angle C$ is: 1

(a) 65° **(b)** 85° **(c)** 95° **(d)** 75°

4. $\frac{2}{3} \times \frac{3}{4} \div \frac{1}{2}$ equals: 1

(a) 1 **(b)** $1\frac{1}{2}$ **(c)** 2 **(d)** $\frac{1}{2}$

5. The sum of the largest 6-digit number and the smallest 7-digit number is: 1

(a) 19,99,998 **(b)** 19,99,999 **(c)** 20,00,001 **(d)** 20,00,000

6. Fill in the number: $6 \times (? + 4) = 78$. The missing number is: 1

(a) 11 **(b)** 8 **(c)** 9 **(d)** 10

7. 7.5 litres of milk is shared equally among 6 persons. Each person gets (in litres): 1

(a) 1.25 **(b)** 2.25 **(c)** 1.35 **(d)** 1.15

8. Subtract $3x + 2$ from $8x + 9$: 1

(a) $5x + 7$ **(b)** $5x + 11$ **(c)** $-5x - 7$ **(d)** $11x + 11$

9. Four angles around a point are 75°, 105°, 90° and x°. Then $x =$ 1

(a) 110° **(b)** 80° **(c)** 90° **(d)** 100°

10. The number of 2-digit palindromes is: 1

(a) 8 **(b)** 9 **(c)** 10 **(d)** 11

11. Two sides of a triangle are 4 cm and 9 cm. The number of possible whole-number lengths of the third side is: 1

(a) 9 **(b)** 8 **(c)** 7 **(d)** 6

12. If $\frac{2}{3}$ of a number is 14, the number is: 1

(a) 42 **(b)** 21 **(c)** $9\frac{1}{3}$ **(d)** $10\frac{1}{2}$

13. Estimate 3,45,678 × 6 by rounding 3,45,678 to the nearest lakh and 6 as given: 1

(a) 1800000 **(b)** 1700000 **(c)** 1900000 **(d)** 1810000

14. A hall has 15 rows with 24 seats each. 35 seats are reserved. Seats available: 1

(a) 335 **(b)** 324 **(c)** 325 **(d)** 326

15. The value of $20 - 2.5 \times 3.2$ is: 1

(a) 12.1 **(b)** 12 **(c)** 11.9 **(d)** 13

16. The nth term of a pattern is $4n + 3$. Its 10th term is: 1

(a) 45 **(b)** 43 **(c)** 44 **(d)** 42

17. An angle is 30° less than its supplement. The angle is: 1

(a) 85° **(b)** 65° **(c)** 95° **(d)** 75°

18. The sum of 5 consecutive numbers starting at 10 is: 1

(a) 62 **(b)** 59 **(c)** 61 **(d)** 60

19. The angles of a triangle are $(60 - d)^\circ$, 60° and $(60 + d)^\circ$. When $d = 10$, the largest angle is: 1

(a) 80° **(b)** 90° **(c)** 70° **(d)** 60°

20. A tap fills $\frac{2}{5}$ of a tank in 1 hour. The time (hours) to fill the whole tank is: 1

(a) $2\frac{1}{2}$ **(b)** $2\frac{5}{6}$ **(c)** $1\frac{1}{4}$ **(d)** 5

SECTION — B (Very Short Answer Type) — 6 × 2 = 12

Answer in about 20–30 words each.

21. An isosceles triangle has perimeter 26 cm and one side 6 cm. Find its other sides (check which case is possible). 2

22. Simplify $\frac{2}{3} \times \frac{9}{10} \div \frac{3}{5}$. 2

23. Find 8,64,00,000 ÷ 24. 2

24. Write an expression: 'the product of 6 and the sum of 9 and 8'. Evaluate it. 2

25. Find $(1.2 \times 0.5) \div 0.04$. 2

26. Subtract $3x - 5$ from $9x + 2$. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. Round 34,56,789 to the nearest 3

(i) thousand

(ii) lakh and find the difference between the two rounded numbers.

28. Find the value of $4 \times (9 + 6) \times (10 - 7)$ by doing the work in steps, and then check by changing the order of two multiplications. 3

29. A pen costs ₹7.5. Riya buys 8 pens and pays ₹100. Find the cost of the pens and the balance she gets back. 3

30. A row of n triangles made with matchsticks needs $2n + 1$ sticks. How many sticks are needed for 9 triangles and for 14 triangles? 3

31. An angle is 6° more than twice its complement. Find the angle and its complement. 3

OR

The sum of a two-digit number and its reverse is 99, and its digits differ by 3. Find the number (tens digit larger). 3

32. In $\triangle PQR$, $PQ = PR$, $\angle Q = (3x - 10)^\circ$ and $\angle R = (x + 6)^\circ$. Find x, all three angles and the exterior angle at P. 3

OR

Mohan has ₹240. He spends $\frac{1}{4}$ of it and shares the rest equally among 6 children. How much does each child get? 3

SECTION — D (Long Answer Type) 6 × 5 = 30

Answer in about 100–150 words each.

33. The expression $3 \times (120 + 85) - 150$ models a situation. Evaluate it, and then write a story problem for it (one line) and give the answer with units. 5

34. A tank of capacity 90 litres is filled by a tap at 7.5 litres per minute. How many minutes does it take to fill the empty tank? How much water is in it after 6 minutes and how much more is needed? 5

35. Raju is r years old. His mother's age is 4 more than 3 times his age. After 5 years, write the ages of both and the sum of their ages. Evaluate the sum for $r = 10$. 5

36. The four angles around a point are in the ratio $1 : 2 : 3 : 4$. Find the angles and say which of them are acute, right or obtuse. 5

OR

List all three-digit palindromes with first digit 1 that are divisible by 3. 5

37. Four sticks of lengths 2 cm, 4 cm, 6 cm and 7 cm are available. Choosing any three, how many different triangles can be made? Test each choice. 5

OR

A jug holds $2\frac{1}{2}$ litres of squash. $\frac{2}{5}$ of it is syrup and the rest is water. Find the syrup and the water. How many glasses of $\frac{1}{4}$ litre can be filled from the jug? 5

38. A company earned ₹24,00,000 in the first year, and in each next year the earning increased by ₹2,50,000. Find its earnings in the 2nd, 3rd and 4th years and the total of the four years. 5

OR

A school buys 30 chairs at ₹600 each and 8 tables at ₹1500 each. A discount of ₹100 per chair is given. Write an expression for the amount payable and find it. 5

END OF QUESTION PAPER Total: 80 Marks

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